

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{3,2,0}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{4,2,-1}[\tilde{\mu}, m_1] - \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 y_d^{i1i2} \right. \\
& \text{LF}_{3,2,0}[\tilde{\mu}, m_2] + 2 g_2^4 m_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{4,2,-1}[\tilde{\mu}, m_2] + \frac{1}{3} g_1^4 m_1^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,2,1,0}[m_1, \tilde{\mu}, m_d^{i2}] + \\
& \frac{1}{6} g_1^4 m_1^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,2,1,0}[m_1, \tilde{\mu}, m_q^{i1}] - 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_1] + \\
& 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] + \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_q^{i1}] - \\
& m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{3,1,1,0}[\tilde{\mu}, m_1, m_2] + \frac{3}{2} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
& 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] + \frac{1}{9} g_1^4 m_1^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,1,1,1,0}[m_1, m_d^{i2}, m_q^{i1}, \tilde{\mu}] + \\
& \frac{3}{4} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} y_d^{i1i2} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_d^{\text{r}}, m_q^{\text{p}}, m_d^{\text{t}}, m_q^{\text{s}}] - \\
& \frac{3}{4} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} y_d^{i1i2} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_d^{\text{r}}, m_q^{\text{s}}, m_d^{\text{t}}, m_q^{\text{p}}] + \\
& \frac{1}{4} \tilde{\mu}^2 y_d^{i1i2} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_e^{\text{r}}, m_l^{\text{p}}, m_e^{\text{t}}, m_l^{\text{s}}] - \\
& \frac{1}{4} \tilde{\mu}^2 y_d^{i1i2} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_e^{\text{r}}, m_l^{\text{s}}, m_e^{\text{t}}, m_l^{\text{p}}] + \\
& \frac{1}{2} \tilde{\mu}^2 y_d^{i1i2} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_l^{\text{p}}, m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{s}}] - \\
& \frac{1}{2} \tilde{\mu}^2 y_d^{i1i2} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_l^{\text{p}}, m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{s}}] + \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} y_d^{i1i2} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_q^{\text{p}}, m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{s}}] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} y_d^{i1i2} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_q^{\text{p}}, m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{s}}] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{\text{sr}} y_d^{i1i2} y_u^{\text{pt}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_q^{\text{p}}, m_u^{\text{r}}, m_q^{\text{s}}, m_u^{\text{t}}] + \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,1,1,1,0}[m_u^{\text{r}}, m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{t}}] - \\
& \frac{3}{2} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{3,1,1,1,-1}[m_u^{\text{r}}, m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{t}}] - \\
& \frac{3}{4} \tilde{\mu}^2 y_d^{i1i2} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,2,1,1,-1}[m_u^{\text{r}}, m_u^{\text{t}}, m_q^{\text{p}}, m_q^{\text{s}}] + \\
& \frac{1}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,1,1,1,0}[\tilde{\mu}, m_1, m_2, m_d^{i2}] + \\
& \frac{2}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{2,1,1,1,0}[\tilde{\mu}, m_1, m_2, m_q^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,1,0}[\tilde{\mu}, m_1, m_d^{\text{r}}, m_q^{\text{p}}] - \\
& \frac{1}{2} m_2 g_2^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,1,0}[\tilde{\mu}, m_2, m_d^{\text{r}}, m_q^{\text{p}}] - \\
& m_2 g_2^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_u^{\text{i}1\text{r}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,1,0}[\tilde{\mu}, m_2, m_q^{\text{p}}, m_u^{\text{r}}] + \\
& \frac{1}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_d^{i1i2} \text{LF}_{1,1,1,1,1,0}[m_1, m_2, m_d^{i2}, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{6} m_1 g_1^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_1, m_d^{\text{r}}, m_q^{i1}, m_q^{\text{p}}, \tilde{\mu}] - \\
& \frac{1}{3} m_1 g_1^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_1, m_d^{i2}, m_d^{\text{r}}, m_q^{\text{p}}, \tilde{\mu}] - \\
& \frac{1}{9} m_1 g_1^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_1, m_d^{i2}, m_d^{\text{r}}, m_q^{i1}, m_q^{\text{p}}] - \\
& \frac{1}{2} m_2 g_2^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_2, m_d^{\text{r}}, m_q^{i1}, m_q^{\text{p}}, \tilde{\mu}] - \\
& m_2 g_2^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_u^{\text{i}1\text{r}} \overline{a_u^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_2, m_q^{i1}, m_q^{\text{p}}, m_u^{\text{r}}, \tilde{\mu}] + \\
& \frac{8}{3} m_3 g_3^2 \tilde{\mu}^2 y_d^{\text{pi}2} y_d^{\text{i}1\text{r}} \overline{a_d^{\text{pr}}} \text{LF}_{1,1,1,1,1,0}[m_3, m_d^{i2}, m_d^{\text{r}}, m_q^{i1}, m_q^{\text{p}}] + \\
& \left. \tilde{\mu}^2 y_d^{\text{pi}2} y_u^{\text{sr}} y_u^{\text{i}1\text{t}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{1,1,1,1,1,0}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}, \tilde{\mu}] \right)
\end{aligned}$$